

Assignment – 6

Q1. Write a program to count the number of times a character appears in a given string.

```
string = input("Enter a string: ")
freq = {}

for ch in string:
    freq[ch] = freq.get(ch, 0) + 1

print("Frequency of every character is:", freq)
```

Output:

Enter a string: Python is easy to learn and Python is powerful. Learning Python is fun.

Frequency of every character is: {'P': 3, 'y': 4, 't': 4, 'h': 3, 'o': 5, 'n': 8, ' ': 12, 'i': 4, 's': 4, 'e': 4, 'a': 4, 'l': 2, 'r': 3, 'd': 1, 'p': 1, 'w': 1, 'f': 2, 'u': 2, '.': 2, 'L': 1, 'g': 1}

Q2. Write a program to convert a number entered by the user into its corresponding number in words. For example, if the input is 876 then the output should be 'Eight Seven Six'. (Hint. use dictionary for keys 0-9 and their values as equivalent words.)

```
MAPPING = {
    0: "Zero",
    1: "One",
    2: "Two",
    3: "Three",
    4: "Four",
    5: "Five",
    6: "Six",
    7: "Seven",
    8: "Eight",
    9: "Nine",
}

num = input("Enter a number: ")
print("The equivalent number is:", end=' ')

for ch in num:
    print(MAPPING[int(ch)], end=' ')
```

Output:

Enter a number: 876

The equivalent number is: Eight Seven Six

Q3. Create a dictionary whose keys are month names and whose values are the number of days in the corresponding months.

- (a) Ask the user to enter a month name and use the dictionary to tell how many days are in the month.
- (b) Print out all of the keys in alphabetical order.

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(c) Print out all of the months with 31 days.

(d) Print out the (key-value) pairs sorted by the number of days in each month.

```
MAPPING = {
    "January": 31,
    "February": 28,
    "March": 31,
    "April": 30,
    "May": 31,
    "June": 30,
    "July": 31,
    "August": 31,
    "September": 30,
    "October": 31,
    "November": 30,
    "December": 31,
}

name = input("Enter a month name: ").title()
print("(a) Number of days in", name, "are:", MAPPING[name])
print("(b) Keys in alphabetical order:", sorted(MAPPING.keys()))
print("(c) Months with 31 days are:", [name for name, days in MAPPING.items() if days == 31])
print("(d) key-value pairs stored by number of days: ", sorted(MAPPING.items(), key=lambda x: x[1]))
```

Output:

```
Enter a month name: February
(a) Number of days in February are: 28
(b) Keys in alphabetical order: ['April', 'August', 'December', 'February', 'January', 'July', 'June', 'March', 'May', 'November', 'October', 'September']
(c) Months with 31 days are: ['January', 'March', 'May', 'July', 'August', 'October', 'December']
(d) key-value pairs stored by number of days: [('February', 28), ('April', 30), ('June', 30), ('September', 30), ('November', 30), ('January', 31), ('March', 31), ('May', 31), ('July', 31), ('August', 31), ('October', 31), ('December', 31)]
```

Q4. Given the dictionary $x = \{ 'k1': 'v1', 'k2': 'v2', 'k3': 'v3' \}$, create a dictionary with the opposite mapping, i.e., write a program to create the dictionary as : $\text{inverted_x} = \{ 'v1': 'k1', 'v2': 'k2', 'v3': 'k3' \}$

```
x = eval(input("Enter a dictionary: "))
inverted_x = {value: key for key, value in x.items()}
print("Inverted dict:", inverted_x)
```

Output:

```
Enter a dictionary: {'k1': 'v1', 'k2': 'v2', 'k3': 'v3'}
Inverted dict: {'v1': 'k1', 'v2': 'k2', 'v3': 'k3'}
```

Q5. Given two dictionaries say D1 and D2. Write a program that lists the overlapping keys of the two dictionaries, i.e., if a key of D1 is also a key of D2, then list it.

```
d1 = eval(input("Enter a dictionary: "))
d2 = eval(input("Enter another dictionary: "))
print("Overlapping keys are:", *(key for key in d1 if key in d2))
```

Output:

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Enter a dictionary: {"a": 10, "b": 20, "c": 30, "d": 40}

Enter another dictionary: {"b": 50, "c": 60, "e": 70}

Overlapping keys are: b c

Q6. Write a program that checks if two same values in a dictionary have different keys. That is, for dictionary D1 = { 'a' : 10, 'b' : 20, 'c' : 10}, the program should print 2 keys have same values and for dictionary D2 = { 'a' : 10, 'b' : 20, 'c' : 30}, the program should print No keys have same values.

```
d1 = eval(input("Enter a dictionary: "))
found = False
seen = {}

for value in d1.values():
    seen[value] = seen.get(value, 0) + 1

for key, value in seen.items():
    if value > 1:
        if not found:
            print("No. of keys with same value are:")

            found = True
            print("Value", value, "repeated", key, "times")
        if not found:
            print("No key have repeated values")
```

Output:

Enter a dictionary: {'a': 10, 'b': 20, 'c': 10}

No. of keys with same value are:

Value 2 repeated 10 times

Q7. Write a program to check if a dictionary is contained in another dictionary e.g., if

d1 = {1:11, 2:12}

d2 = {1:11, 2:12, 3:13, 4:15}

then d1 is contained in d2.

```
d1 = eval(input("Enter a dictionary: "))
d2 = eval(input("Enter another dictionary: "))
flag = True

for key, value in d1.items():
    if key not in d2 or d2[key] != value:
        flag = False

if flag:
    print("d2 contains d1")

flag = True

for key, value in d2.items():
    if key not in d1 or d1[key] != value:
```

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```

flag = False

if flag:
    print("d1 contains d2")

```

Output:

```

Enter a dictionary: {1: 11, 2: 12}
Enter another dictionary: {1: 11, 2: 12, 3: 13, 4: 15}
d2 contains d1

```

Q8. A dictionary D1 has values in the form of lists of numbers. Write a program to create a new dictionary D2 having same keys as D1 but values as the sum of the list elements e.g.,

D1 = {'A' : [1, 2, 3] , 'B' : [4, 5, 6]}

then

D2 is {'A' :6, 'B' : 15}

```

d1 = eval(input("Enter a dictionary: "))
d2 = {key: sum(value) for key, value in d1.items()}
print("Resultant dict:", d2)

```

Output:

```

Enter a dictionary: {'A': [1, 2, 3], 'B': [4, 5, 6]}
Resultant dict: {'A': 6, 'B': 15}

```

Q9. Create a dictionary 'ODD' of odd numbers between 1 and 10, where the key is the decimal number and the value is the corresponding number in words. Perform the following operations on this dictionary:

- (a) Display the keys**
- (b) Display the values**
- (c) Display the items**
- (d) Find the length of the dictionary**
- (e) Check if 7 is present or not**
- (f) Check if 2 is present or not**
- (g) Retrieve the value corresponding to the key 9**
- (h) Delete the item from the dictionary corresponding to the key 9**

```

ODD = {
    1: "One",
    3: "Three",
    5: "Five",
    7: "Seven",
    9: "Nine",
}

print("(a) keys:", ODD.keys())
print("(b) values:", ODD.values())
print("(c) items:", ODD.items())
print("(d) length:", len(ODD))
print("(e) Is 7 present in dict?", 7 in ODD)
print("(f) Is 2 present in dict?", 2 in ODD)
print("(g) Value of 9:", ODD[9])
del ODD[9]
print("(h) Dict after deleting key 9:", ODD)

```

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Output:

```
(a) keys: dict_keys([1, 3, 5, 7, 9])
(b) values: dict_values(['One', 'Three', 'Five', 'Seven', 'Nine'])
(c) items: dict_items([(1, 'One'), (3, 'Three'), (5, 'Five'), (7, 'Seven'), (9, 'Nine')])
(d) length: 5
(e) Is 7 present in dict? True
(f) Is 2 present in dict? False
(g) Value of 9: Nine
(h) Dict after deleting key 9: {1: 'One', 3: 'Three', 5: 'Five', 7: 'Seven'}
```

Q10. Write a program to input your friends' names and their Phone Numbers and store them in the dictionary as the key-value pair. Perform the following operations on the dictionary:

- a) Display the name and phone number of all your friends**
- b) Add a new key-value pair in this dictionary and display the modified dictionary**
- c) Delete a particular friend from the dictionary**
- d) Modify the phone number of an existing friend**
- e) Check if a friend is present in the dictionary or not**
- f) Display the dictionary in sorted order of names**

```
dict1 = eval(input("Enter the dictionary of names and phone nos.: "))
print("(a) All names and phone nos.:", dict1)
print()

element = eval(input("Enter a new key-value tuple: "))
dict1[element[0]] = element[1]
print("(b) Dict after adding", element, ":", dict1)
print()

name = input("Enter a name to delete from the dictionary: ")
del dict1[name]
print("(c) Dict after removing", name, ":", dict1)
print()

element = eval(input("Enter the name and updated phone no. tuple: "))
dict1[element[0]] = element[1]
print("(d) Dict after updating", element[0], "with", element[1], ":", dict1)
print()

name = input("Enter a name to check if it is present or not: ")
print("(e) Is", name, "present in the dictionary?", name in dict1)
print()

print("(f) Dictionary in sorted order:", sorted(dict1.items()))
```

Output:

```
Enter the dictionary of names and phone nos.: {"Amit": 9876543210, "Riya": 9123456780, "Kunal": 9988776655}
(a) All names and phone nos.: {'Amit': 9876543210, 'Riya': 9123456780, 'Kunal': 9988776655}

Enter a new key-value tuple: ("Neha", 9012345678)
(b) Dict after adding ('Neha', 9012345678) : {'Amit': 9876543210, 'Riya': 9123456780, 'Kunal': 9988776655, 'Neha': 9012345678}
```

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Enter a name to delete from the dictionary: Kunal

(c) Dict after removing Kunal : {'Amit': 9876543210, 'Riya': 9123456780, 'Neha': 9012345678}

Enter the name and updated phone no. tuple: ("Riya", 9000000000)

(d) Dict after updating Riya with 9000000000 : {'Amit': 9876543210, 'Riya': 9000000000, 'Neha': 9012345678}

Enter a name to check if it is present or not: Amit

(e) Is Amit present in the dictionary? True

(f) Dictionary in sorted order: [('Amit', 9876543210), ('Neha', 9012345678), ('Riya', 9000000000)]

Q11. Word Frequency in a Paragraph

```
paragraph = input("Enter a paragraph: ")
```

```
freq = {}
```

```
for word in paragraph.split():
```

```
    freq[word] = freq.get(word, 0) + 1
```

```
print("Word Frequency:", freq)
```

Output:

Enter a paragraph: Python is easy to learn and Python is powerful. Learning Python is fun.

Word Frequency: {'Python': 3, 'is': 3, 'easy': 1, 'to': 1, 'learn': 1, 'and': 1, 'powerful.': 1, 'Learning': 1, 'fun.': 1}

Q12. Find Common Keys in Two Dictionaries

```
d1 = eval(input("Enter a dictionary: "))
```

```
d2 = eval(input("Enter another dictionary: "))
```

```
print("Common Keys are:", *(key for key in d1 if key in d2))
```

Output:

Enter a dictionary: {"a": 10, "b": 20, "c": 30, "d": 40}

Enter another dictionary: {"b": 50, "c": 60, "e": 70}

Common Keys are: b c

Q13. Invert Dictionary with Duplicate Value

Input: d = {'a':1, 'b':2, 'c':1, 'd':3, 'e':2}

Output: {1: ['a', 'c'], 2: ['b', 'e'], 3: ['d']}

```
d = eval(input("Enter a dictionary: "))
```

```
res = {}
```

```
for key, value in d.items():
```

```
    res[value] = res.get(value, [])
```

```
    res[value].append(key)
```

```
print("resultant dict:", res)
```

Output:

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Enter a dictionary: {'a': 1, 'b': 2, 'c': 1, 'd': 3, 'e': 2}
 resultant dict: {1: ['a', 'c'], 2: ['b', 'e'], 3: ['d']}

Q14. Using dictionary, print the first character that does not repeat.

Input: "aabbccddeefg"

Output: f

```
string = input("Enter a string: ")
freq = {}

for ch in string:
    freq[ch] = freq.get(ch, 0) + 1

for ch in string:
    if freq[ch] == 1:
        print("The first character that does not repeat:", ch)
        break
```

Output:

Enter a string: "aabbccddeefg"
 The first character that does not repeat: f

Q15. Using dictionary, count how many times each number appears.

```
lst = list(eval(input("Enter the list of numbers: ")))
freq = {}
```

```
for num in lst:
    freq[num] = freq.get(num, 0) + 1

print("Frequency of each number is:", freq)
```

Output:

Enter the list of numbers: [1, 2, 3, 2, 4, 1, 2, 5, 3, 3, 3]
 Frequency of each number is: {1: 2, 2: 3, 3: 4, 4: 1, 5: 1}

Q16. Create a dictionary where keys are word lengths and values are lists of words.

Input: ["cat", "dog", "elephant", "rat", "tiger"]

Output: {3:['cat', 'dog', 'rat'], 5:['tiger'], 8:['elephant']}

```
words = list(eval(input("Enter the list of strings: ")))
dict1 = {}

for word in words:
    dict1[len(word)] = dict1.get(len(word), [])
    dict1[len(word)].append(word)

print("Resultant dict is:", dict1)
```

Output:

Enter the list of strings: ["cat", "dog", "elephant", "rat", "tiger"]
 Resultant dict is: {3: ['cat', 'dog', 'rat'], 8: ['elephant'], 5: ['tiger']}

Q17. Sort dictionary in ascending order of values.

```
dict1 = eval(input("Enter a dictionary: "))  
print(dict(sorted(dict1.items(), key=lambda x: x[1])))
```

Output:

```
Enter a dictionary: {"Amit": 9876543210, "Riya": 9123456780, "Kunal": 9988776655}  
{'Riya': 9123456780, 'Amit': 9876543210, 'Kunal': 9988776655}Count is: 5
```